

Fall 2024 ECEN 629 Homework Set 5

Prob. 1 In many cases, we may need to compute the projection of a point onto a convex set (e.g., in projected gradient descent). The projection of x_0 on convex set is defined as the optimal point for the following optimization problem:

$$\begin{aligned} & \text{minimize: } \|x - x_0\| \\ & \text{subject to: } x \in \mathcal{C}. \end{aligned}$$

This projection is sometimes easy, sometimes hard. For the easy cases, analytical solutions can be found.

- (a) Suppose the norm we use is the ℓ_∞ norm, and the set $\mathcal{C}$ is $\{x \mid \|x\|_2 \leq 1\}$. Find the projection of x_0 .
- (b) Suppose the norm we use is the ℓ_2 norm, and the set $\mathcal{C}$ is $\{x \mid \|x\|_2 \leq 1\}$. Find the projection of x_0 .
- (c) Suppose the norm we use is the ℓ_∞ norm, and the set $\mathcal{C}$ is $\{x \mid l \preceq x \preceq u\}$. Find the projection of x_0 .
- (d) Suppose the norm we use is the ℓ_2 norm, and the set $\mathcal{C}$ is $\{x \mid l \preceq x \preceq u\}$. Find the projection of x_0 .
- (e) Suppose the norm we use is the ℓ_1 norm, and the set $\mathcal{C}$ is $\{x \mid l \preceq x \preceq u\}$. Find the projection of x_0 .
- (f) Suppose the norm we use is the ℓ_2 norm, and the set $\mathcal{C}$ is $\{x \mid Ax = b\}$. Find the projection of x_0 . Assume the matrix A has full row rank, and has more columns than rows.
- (g) Provide one example of a convex set $\mathcal{C}$ that this projection is not easy to obtain analytical solutions or to compute.

Solution:

- (a) The problem has a geometric interpretation, but we can write the optimization problem as

$$\begin{aligned} & \text{minimize: } t \\ & \text{subject to: } -t \cdot \vec{1} \preceq x - x_0 \preceq t \cdot \vec{1} \\ & \quad \quad \quad x^T x \leq 1. \end{aligned}$$

We can use the KKT condition (strong duality clearly holds) to find the optimal solution. We write the Lagrangian as

$$L = t + \lambda_1^T (-t \cdot \vec{1} - x + x_0) + \lambda_2^T (-t \cdot \vec{1} + x - x_0) + \lambda_3(x^T x - 1).$$

The vanishing gradient condition is:

$$\begin{aligned} -\lambda_1 + \lambda_2 + 2\lambda_3 x &= 0 \\ 1 - \lambda_1^T \cdot \vec{1} - \lambda_2^T \cdot \vec{1} &= 0. \end{aligned}$$

We consider several cases:

- i. If $\lambda_3 = 0$, then $\lambda_1 = \lambda_2$. If for any i , $\lambda_{1,i} = \lambda_{2,i} > 0$, then by the complementary slackness we must have $t = 0$, and as a consequence, $x = x_0$. On the other hand, $\lambda_{1,i} = \lambda_{2,i} = 0$ for all i cannot occur due to the second condition above. Thus this case only occurs when $\|x_0\|_2 \leq 1$, i.e., x_0 is in the set C .
- ii. If $\lambda_3 > 0$, then we must have $x^T x = 1$, i.e., x^* is on the norm ball surface. Let us consider a few different cases:
 - A. $\lambda_{1,i} = \lambda_{2,i} > 0$ for some i , then again we must have $t = 0$. This case only occurs when $\|x_0\|_2 = 1$, and $x^* = x_0$.
 - B. $\lambda_{1,i} = \lambda_{2,i} = 0$ for some i , then $x_i^* = 0$. For other i 's, we have $|x_i^* - x_{0,i}| = t$. In other words, either $x_i^* = 0 \in [x_{0,i} - t, x_{0,i} + t]$, or $|x_i^* - x_{0,i}| = t$. Now without loss of generality, assume $|x_{0,1}| \geq |x_{0,2}| \geq \dots \geq |x_{0,n}|$. The solution structure implies that we can search for the integer k such that $\sum_{i=1}^k (x_{0,i} - t \cdot \text{sgn}(x_{0,i}))^2 = 1$ for which the solution t^* satisfy $0 \in [x_{0,k+1} - t^*, x_{0,k+1} + t^*]$. It is easy to see there is only one solution by the monotonicity of the solution t with respect to k . A naive approach would be to solve the quadratic equation given above for a total of n times, but better approaches exist (we actually can write down the solution explicitly) to make the computation even faster.
- (b) We have seen this many times, and the solutions is: if $\|x_0\|_2 \leq 1$, then $x^* = x_0$; otherwise, $x^* = x_0 / \|x_0\|_2$.
- (c) Notice that there is essentially no tradeoff among the components of x , and each x_i component can be assigned independently from other components. Therefore, a solution x^* is as follows

$$x_i^* = \begin{cases} l_i & (x_0)_i \leq l_i \\ u_i & (x_0)_i \geq u_i \\ (x_0)_i & \text{otherwise.} \end{cases}$$

Note that there may be multiple solutions: if $|x_i^* - x_{0,i}| < \|x^* - x_0\|_\infty \triangleq \tau$, then we can choose x_i^* differently as long as $|x_i^* - x_{0,i}| \leq \tau$.

- (d) The solution x^* is the same as the previous case.
- (e) This has the same same solution as the previous two cases.
- (f) We can write the optimization problem

$$\begin{aligned} & \text{minimize: } x^T x \\ & \text{subject to: } Ax - b = 0. \end{aligned}$$

This is a convex problem, and (the general) Slater's condition holds. Therefore, we can solve the KKT condition to find the optimal solution. From the vanishing gradient condition, we obtain

$$x = -\frac{1}{2}A^T \nu. \tag{1}$$

Because the equality constraint $Ax = b$, we obtain that

$$-AA^T \nu - 2b = 0. \tag{2}$$

which gives $\nu = -2(AA^T)^{-1}b$, where we used the fact A has full row rank, thus (AA^T) is invertible. It follows that $x^* = A^T(AA^T)^{-1}b$.

- (g) When the set is specified by more general constraint sets $f_i(x) \leq 0$, $i = 1, 2, \dots, m$, and the norm is not of the special forms, this project is not easy to compute. For example, let $f_i(x) = x^T Q_i x + p_i x + c_i$, where Q_i is a positive semidefinite matrix, and the norm is ℓ_4 norm, then the problem becomes more difficult.

Prob. 2 Consider the problem:

$$\begin{aligned} & \text{minimize: } \text{length}(x) \\ & \text{subject to: } \|Ax - b\| \leq \epsilon, \end{aligned}$$

where the length function $\text{length}(x) = \min\{k | x_i = 0 \text{ for } i > k\}$. The decision variables $x \in \mathbb{R}^n$, and A, b, ϵ are fixed parameters.

- (a) Show the problem is not convex;
- (b) Show the problem is quasiconvex;
- (c) Is the length function equivalent to the sparsity, i.e., the number of non-zero elements of x ?
- (d) (optional) Formulate a problem where both the sparsity and the length of x are to be minimized. Propose an approach to solve this problem efficiently.

Solution:

- (a) We only need a counter example: Let $x = (1, 0)^T$ and $y = (1, 1)^T$, then $0.5\text{length}(x) + 0.5\text{length}(y) = 1.5$ however $\text{length}(0.5x + 0.5y) = \text{length}(x)(1, 0.5) = 2$.
- (b) To show it is quasiconvex, we only need to show that the set $\{x | \text{length}(x) \leq \alpha\}$ is convex. The condition $\text{length}(x) \leq \alpha$ is equivalent to the condition $x_k = 0$ for all $k > \alpha$. Take any two vectors x, y such that $x_k = y_k = 0$ for all $k > \alpha$, clearly their convex combination also has this property. Therefore, this set is indeed convex.
- (c) No this is not equivalent to the sparsity. When minimizing the length, we seek to find a small number k of non-zero elements, however, the locations are fixed to the first k . In contrast, for sparsity, the non-zero elements can be at any locations, not necessarily the first k .

Prob. 3 Suppose the measurement noise of a scalar quantity x has the truncated Gaussian distribution with parameter $a > 0 > b$, such that the observed measurement y is distributed according to the following distribution

$$p_x(y) = \frac{1}{\Phi \sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y-x)^2}{2\sigma^2} \right], \quad \text{for } b \leq y - x \leq a,$$

and $p_x(y) = 0$ otherwise, where Φ is a constant, and σ^2 is a fixed and known value.

- (a) Write down an expression for Φ .
- (b) Write down the ML estimation problem, when we have m iid scalar samples $\{y_i\}$, as an optimization problem; simplify it into a standard convex optimization form.
- (c) Suppose $m = 8$, $(y_1, y_2, \dots, y_8) = (-1, 3, -6, 6, 5, 5, 8, 7)$, $\sigma^2 = 1$, and $a = 7$ and $b = -10$. What is the solution to your optimization problem?

Solution:

- (a) It is essentially the normalization factor:

$$\Phi = \int_b^a \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{z^2}{2\sigma^2} \right] dz.$$

- (b) The log likelihood function is

$$\log \prod_i^m p_x(y_i) = \begin{cases} m \log \frac{1}{\Phi\sqrt{2\pi\sigma}} - \sum_{i=1}^m \frac{(y_i-x)^2}{2\sigma^2} & x+a \geq y_i \geq x+b, i=1,2,\dots,m \\ -\infty & \text{otherwise.} \end{cases}$$

The ML estimation problem is thus equivalent to

$$\begin{aligned} & \text{minimize: } mx^2 - 2\left(\sum_{i=1}^m y_i\right)x \\ & \text{subject to: } x \leq y_i - b, \quad i = 1, 2, \dots, m \\ & \quad \quad \quad x \geq y_i - a, \quad i = 1, 2, \dots, m. \end{aligned}$$

Note that x is the variable here, and y_i 's are known.

- (c) The objective function is quadratic in x , and the minimum is $\bar{x} = (\sum_{i=1}^m y_i)/m$, if the constraints were not present. However, with this constraint, we need to take into consideration of the feasible set of x . This condition can be rewritten as $x \in [\max(y_i) - a, \min(y_i) - b]$, and thus with the given values the feasible condition is equivalent to $4 \geq x \geq 1$. Here $\bar{x} = 27/8$, thus $x^* = 27/8$.

Prob. 4 Write a simple supporting vector machine program using Matlab/CVX/Python as follows:

- (a) Write a function to generate $n = 800$ random points in the dimensional space from the Gaussian mixture distribution

$$\begin{aligned} P(x|\mu_{1,x}, \mu_{1,y}, \sigma_{1,x}^2, \sigma_{1,y}^2, \pi_1, \mu_{2,x}, \mu_{2,y}, \sigma_{2,x}^2, \sigma_{2,y}^2, \pi_2) \\ = \frac{\pi_1}{2\pi\sigma_{1,x}\sigma_{1,y}} \exp \left(-\frac{(x-\mu_{1,x})^2}{2\sigma_{1,x}^2} - \frac{(y-\mu_{1,y})^2}{2\sigma_{1,y}^2} \right) \\ + \frac{\pi_2}{2\pi\sigma_{2,x}\sigma_{2,y}} \exp \left(-\frac{(x-\mu_{2,x})^2}{2\sigma_{2,x}^2} - \frac{(y-\mu_{2,y})^2}{2\sigma_{2,y}^2} \right) \end{aligned}$$

where $(\mu_{1,x}, \mu_{1,y}) = (0, 0)$, $(\mu_{2,x}, \mu_{2,y}) = (2, 1)$, $(\sigma_{1,x}^2, \sigma_{1,y}^2) = (1, 0.5)$, $(\sigma_{2,x}^2, \sigma_{2,y}^2) = (\sqrt{2}, 2)$, $\pi_1 = 0.4$ and $\pi_2 = 0.6$. In other words, in this mixture Gaussian distribution, the modes are chosen at $(0.4, 0.6)$ probability. Draw the scatter plot of these points, and label each point using either a square or a cross, depending on which mode (-1 or 1) the data point was produced from.

- (b) Write a program to call an optimization solver to solve the dual optimization problem, to find the support vector classifier using 600 samples, and plot the classifier and on the scatter plot of these 600 samples.
- (c) Classify the remaining 200 (holdout) points using the classifier you obtained from SVM; plot (in a new figure) the 200 points and classifier result. What is the error rate?
- (d) Complete the task in (b) and (c) using existing Python/Matlab functions or packages. Compare to the result you obtained.

Solution: See additional material.

Prob. 5 Let $\gamma > 1$. The function

$$f(x_1, x_2) = \begin{cases} \sqrt{x_1^2 + \gamma x_2^2} & |x_2| \leq x_1 \\ \frac{x_1 + \gamma |x_2|}{\sqrt{1+\gamma}} & \text{otherwise} \end{cases}$$

is convex.

- (a) Plot the function and visually identify (approximately) its optimal solution;
- (b) Let $\gamma = 100$. Let $x^{(k)} = (1, 5)$ be the point in the k -th iteration. Find the search direction of the gradient descent method.
- (c) Again let $\gamma = 100$, and let $x^{(k)} = (1, 5)$ be the point in the k -th iteration. Find the search direction of the steepest descent method under the ℓ_1 norm;
- (d) Again let $\gamma = 100$, and let $x^{(k)} = (1, 5)$ be the point in the k -th iteration. With the gradient descent method and backtracking line search of parameters $(\alpha, \beta) = (0.45, 0.9)$, will the step size be determined in two iterations in the line search procedure?
- (e) Applying the gradient descent algorithm on this function, with starting point $x^{(0)} = (\gamma, 1)$ with an exact line search. It can be shown that the iterations for $x_1^{(k)}$ is

$$x_1^{(k)} = \gamma \left(\frac{\gamma - 1}{\gamma + 1} \right)^k.$$

Find the value $x_2^{(k)}$ in that corresponding iteration. What does the algorithm converge to? Is it optimal?

Solution:

- (a) This problem is unbounded below.
- (b) We need to compute the gradient here:

$$\nabla f(x) = \begin{cases} \left(\frac{x_1}{\sqrt{x_1^2 + \gamma x_2^2}}, \frac{\gamma x_2}{\sqrt{x_1^2 + \gamma x_2^2}} \right)^T & |x_2| \leq x_1 \\ \left(\frac{1}{\sqrt{1+\gamma}}, \frac{\text{sign}(x_2)\gamma}{\sqrt{1+\gamma}} \right)^T & \text{otherwise} \end{cases} \quad (3)$$

Note at the boundary, the gradient is indeed continuous. Substituting $(x_1, x_2) = (1, 5)$, we obtain

$$\nabla f(x) = \left(\frac{1}{\sqrt{101}}, \frac{100}{\sqrt{101}} \right)^T. \quad (4)$$

The search direction is thus $\Delta x = -\nabla f(x) = \left(-\frac{1}{\sqrt{101}}, -\frac{100}{\sqrt{101}} \right)^T$.

- (c) This is the coordinate descent direction. The largest absolute value in $\nabla f(x)$ is thus for x_2 , and the direction is $\Delta x_{sd} = \|\nabla f(x)\|_\infty (0, -1)^T = \left(0, -\frac{100}{\sqrt{101}} \right)^T$. Here since we only ask about the direction, you can choose to use the unnormalized step Δx_{sd} , as given on textbook page 478.

Note: In Steepest Descent, we usually use Δx_{sd} , instead of Δx_{nsd} . This choice will make a difference here when choosing the stepsize. It can be verified that with the former, we actually choose a much more aggressive step than with the latter.

- (d) Following Algorithm 9.2 in the textbook. We first determine $f(x + \Delta x) \approx 49.35$, and check whether $f(x + \Delta x) > f(x) + 0.45 * 1 * \nabla f(x)^T \Delta x \approx 5.193$. Thus the iteration will not stop. We continue with $t = \beta = 0.9$. In the second step $f(x + 0.9\Delta x) \approx 44.42$ and $f(x) + 0.45 * 0.9 * \nabla f(x)^T \Delta x \approx 9.75$, thus the algorithm will not terminate with $\beta = 0.9$.
- (e) $x_2^{(k)} = \left(-\frac{\gamma-1}{\gamma+1}\right)^k$. Thus the descent algorithm clearly converges to $(0,0)$; this is a stationary point. However this is not the optimal solution, since f is not bounded below. Recall we mentioned that our algorithm had several technical conditions to guarantee convergence to the optimal value. This problem is not strongly convex and it is also not smooth, i.e., the smallest eigenvalue of the Hessian matrix is not always bounded away from zero, and the largest eigenvalue is not finite everywhere. Particularly, the first violation makes the algorithm go awry.

Prob. 6 Consider the optimization problem

$$\begin{aligned} \text{minimize: } & \log \left(\sum_{i=1}^n i e^{x_i} \right) + \sum_{i=1}^n i^2 (x_i - i)^2 \\ \text{subject to: } & \sum_{i=1}^n x_i = P. \end{aligned}$$

- (a) Let $P = 15$. Assume $n = 5$. If the starting point is $x^{(0)} = (1, 2, 3, 4, 5)$, find the gradient, the Hessian matrix, and the first Newton step. Write down the equation, but you can solve for it numerically.
- (b) Let $P = 15$. Assume $n = 50$. Write your own Newton method program (in c/c++ or python) with infeasible start to calculate the optimal solution.
- (c) Use CVX package to solve the same problem in (b).

Solution: We need to compute the gradient and the Hessian

$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right)^T$$

where

$$\frac{\partial f(x)}{\partial x_i} = \frac{i e^{x_i}}{\sum_{k=1}^n k e^{x_k}} + 2i^2 (x_i - i).$$

Similarly for the Hessian matrix H , its elements are

$$H_{i,j} = \begin{cases} \frac{-ij e^{x_i+x_j}}{(\sum_{i=1}^n i e^{x_i})^2} & i \neq j \\ \frac{i e^{x_i}}{\sum_{k=1}^n k e^{x_k}} + \frac{-i^2 e^{x_i}}{(\sum_{k=1}^n k e^{x_k})^2} + 2i^2 & i = j \end{cases}$$

For the point $x = (1, 2, 3, 4, 5)^T$, substituting in the equation above, we have

$$\nabla f(x) = \begin{pmatrix} 0.0026 \\ 0.0142 \\ 0.0580 \\ 0.2104 \\ 0.7148 \end{pmatrix}.$$

$$\nabla^2 f(x) = \begin{pmatrix} 2.0026 & 0 & -0.0002 & -0.006 & -0.0019 \\ 0 & 8.0140 & -0.0008 & -0.0030 & -0.0102 \\ -0.0002 & -0.0008 & 18.0547 & -0.0122 & -0.0415 \\ -0.0006 & -0.0030 & -0.0122 & 32.1661 & -0.1504 \\ -0.0019 & -0.0102 & -0.0415 & -0.1504 & 50.2039 \end{pmatrix}.$$

(a) The Newton step is obtained by

$$\begin{pmatrix} 2.0026 & 0 & -0.0002 & -0.006 & -0.0019 & 1 \\ 0 & 8.0140 & -0.0008 & -0.0030 & -0.0102 & 1 \\ -0.0002 & -0.0008 & 18.0547 & -0.0122 & -0.0415 & 1 \\ -0.0006 & -0.0030 & -0.0122 & 32.1661 & -0.1504 & 1 \\ -0.0019 & -0.0102 & -0.0415 & -0.1504 & 50.2039 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \Delta X_{nt} \\ w \end{pmatrix} = - \begin{pmatrix} 0.0026 \\ 0.0142 \\ 0.0580 \\ 0.2104 \\ 0.7148 \\ 0. \end{pmatrix}$$

Thus we have

$$\Delta X_{nt} = \begin{pmatrix} 0.0173 \\ 0.0029 \\ -0.0012 \\ -0.0054 \\ -0.0135 \end{pmatrix}$$

(b) $p^* = 976958.861045$. See additional material.

(c) You should get the same answer as (b). See additional material.

Prob. 7 Consider the optimization problem (similarly as above but with an inequality constraint)

$$\begin{aligned} \text{minimize: } & \log \left(\sum_{i=1}^n i e^{x_i} \right) + \sum_{i=1}^n i^2 (x_i - i)^2 \\ \text{subject to: } & \sum_{i=1}^n x_i \leq P. \end{aligned}$$

- (a) Let $P = 20$. Assume $n = 5$. If the starting point is $x^{(0)} = (1, 2, 3, 4, 5)$. Find the gradient and the Hessian of the logarithmic barrier function.
- (b) Let $P = 20$. Assume $n = 5$. If the starting point is $x^{(0)} = (1, 2, 3, 4, 5)$. In the log barrier method (when $t = 10$), find the first Newton step. Write down the equation, but you can solve for it numerically.
- (c) Use CVX package to solve the same problem in (b).

Solution:

(a) We have

$$\begin{aligned} \nabla \phi(x) &= (0.2, 0.2, 0.2, 0.2, 0.2)^T \\ \nabla^2 \phi(x) &= \begin{pmatrix} 0.04 & 0.04 & 0.04 & 0.04 & 0.04 \\ 0.04 & 0.04 & 0.04 & 0.04 & 0.04 \\ 0.04 & 0.04 & 0.04 & 0.04 & 0.04 \\ 0.04 & 0.04 & 0.04 & 0.04 & 0.04 \\ 0.04 & 0.04 & 0.04 & 0.04 & 0.04 \end{pmatrix} \end{aligned}$$

(b) We will be minimizing $g(x) = 10f(x) + \phi(x)$.

$$\nabla g(x) = 10\nabla f(x) + \nabla \phi(x) = (0.2262, 0.342, 0.7804, 2.3035, 7.3475)^T.$$

Similarly, we can write down $\nabla^2 g(x) = 10\nabla^2 f(x) + \nabla^2 \phi(x)$. Now we do not have an equality constraint, which gives

$$\Delta X_{nt} = -(\nabla^2 g(x))^{-1} \nabla g(x) = \begin{pmatrix} -0.0112 \\ -0.0043 \\ -0.0044 \\ -0.0072 \\ -0.0147 \end{pmatrix}$$

(c) $p^* \approx 6.939$. See additional material.

Prob. 8 Consider the unconstrained optimization problem

$$\text{minimize: } e^{x_1+2x_2} + x_2^2 + x_3^2 - x_2x_3$$

where $x = (x_1, x_2, x_3)^T \in \mathbf{R}^3$ is the variable. Let $x^{(k)} = (1, 1, 1)$ be the point in the k -th iteration.

(a) Compute the next Newton step.

(b) Suppose the optimization problem has two additional constraints:

$$x_1 + x_2 + x_3 \leq 4$$

$$x_2 - x_3 \leq 2.$$

Compute the next Newton step using the log-barrier method with the same $x^{(k)}$ and the barrier weighting parameter $t = 20$.

Solution:

(a) This is an unconstrained optimization problem. In order to compute the Newton step, we need to compute the gradient and the Hessian.

$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right)^T$$

which gives

$$\frac{\partial f(x)}{\partial x_i} = \begin{pmatrix} e^{x_1+2x_2} \\ 2e^{x_1+2x_2} + 2x_2 - x_3 \\ 2x_3 - x_2 \end{pmatrix}.$$

Similarly for the Hessian matrix

$$\nabla^2 f(x) = \begin{pmatrix} e^{x_1+2x_2} & 2e^{x_1+2x_2} & 0 \\ 2e^{x_1+2x_2} & 4e^{x_1+2x_2} + 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

For the point $x = (1, 1, 1)^T$, you will need to substitute it in the equation above, and solve for the equation

$$\Delta x_{nt} = -\nabla^2 f(x)^{-1} \nabla f(x) = - \begin{pmatrix} 20.0855 & 40.1711 & 0 \\ 40.1711 & 82.3421 & -1.0000 \\ 0 & -1.0000 & 2.0000 \end{pmatrix}^{-1} \begin{pmatrix} 20.0855 \\ 41.1711 \\ 1.0000 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

(b) Here we have

$$\phi(x) = -\log(x_1 + x_2 + x_3 - 4) - \log(x_2 - x_3 - 2),$$

however, to compute the gradient and Hessian of $\phi(x)$, we only need to use the function value, the gradient, and the Hessian of f_i 's. Here

$$\nabla\phi(x) = \frac{1}{1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1.5 \\ 0.5 \end{pmatrix}.$$

Similarly, use the formula given on page 5 of the lecture slides, we can find the Hessian of $\phi(x)$. Following the footsteps in Problem 4(b), you will be able to compute the next Newton step.